REVISION WORKSHEET

UNITED ARAB EMIRATES
MINISTRY OF EDUCATION

الإمارات العربية المتحدة
وزارة التربية والتعليم

AL KHALIDIYA BOYS SCHOOL

Cycle 3 (10048) | Dibba Al Hisn | Sharjah Eastern

Refraction of Light Practice Questions

Grade 10 Advanced | Module 16 Lesson 3 | Refraction of Light

Name: _____ Class: _____ Date: _____

Answer all 30 questions. Select one option for each MCQ. Show equations, working, units and explanations for written questions.

Useful equations and guidance: n = c/v; n1 sin i = n2 sin r; sin c = n2/n1 for n1 > n2. Angles are measured from the normal.

Term 1 2026–2027 | Textbook alignment: uploaded Grade 10 book, printed pages 19–25.

P01. When light enters glass from air, its frequency:

A. Always increases B. Always decreases
C. Becomes zero D. Remains unchanged

P02. Total internal reflection requires light to travel:

A. From higher index to lower index above the critical angle B. From lower index to higher index at any angle
C. Along the normal only D. Through a vacuum only

P03. At the critical angle, the refracted ray is:

A. Along the normal B. At 90° to the normal
C. At the same angle as the incident ray D. Absent because the frequency is zero

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P04. Dispersion by a prism occurs because refractive index depends on:

A. Beam area only B. The observer's height
C. Wavelength D. The prism's mass only

P05. A ray entering a higher-index medium obliquely usually bends:

A. Away from the normal B. Through 90° always
C. Along the boundary always D. Toward the normal

P06. Light travels at 2.14×10^8 m/s in a transparent medium. Calculate its refractive index using $c = 3.00 \times 10^8$ m/s.

A. 2.804 B. 0.7009
C. 1.402 D. -1.402

P07. Light enters glass (n = 1.50) from air (n = 1.00) at 25° to the normal. Find the refraction angle.

A. 32.73 ° B. 8.182 °
C. -16.36 ° D. 16.36 °

P08. Light passes from a medium of index 1.50 toward air (n = 1.00). Calculate its critical angle.

A. 41.81 ° B. 83.62 °
C. 20.91 ° D. -41.81 °

P09. A medium has refractive index 1.40. Calculate light speed in it. Use $c = 3.00 \times 10^8$ m/s.

A. 4.29×10^8 m/s B. 2.14×10^8 m/s
C. 1.07×10^8 m/s D. -2.14×10^8 m/s

P10. Light of wavelength 550 nm in air enters glass of refractive index 1.50. Find its wavelength in glass.

A. 733.3 nm B. 183.3 nm
C. 366.7 nm D. -366.7 nm

P11. Define refraction.

P12. Define refractive index.

P13. Define critical angle.

P14. State both conditions for total internal reflection.

P15. Explain the main cause of a mirage near a hot road.

P16. Light travels at 2×10^8 m/s in a transparent medium. Calculate its refractive index using $c = 3.00 \times 10^8$ m/s.

P17. Light travels at 1.88×10^8 m/s in a transparent medium. Calculate its refractive index using $c = 3.00 \times 10^8$ m/s.

P18. Light enters glass (n = 1.50) from air (n = 1.00) at 30° to the normal. Find the refraction angle.

P19. Light enters glass (n = 1.50) from air (n = 1.00) at 35° to the normal. Find the refraction angle.

P20. Light passes from a medium of index 1.60 toward air (n = 1.00). Calculate its critical angle.

P21. Light passes from a medium of index 1.70 toward air (n = 1.00). Calculate its critical angle.

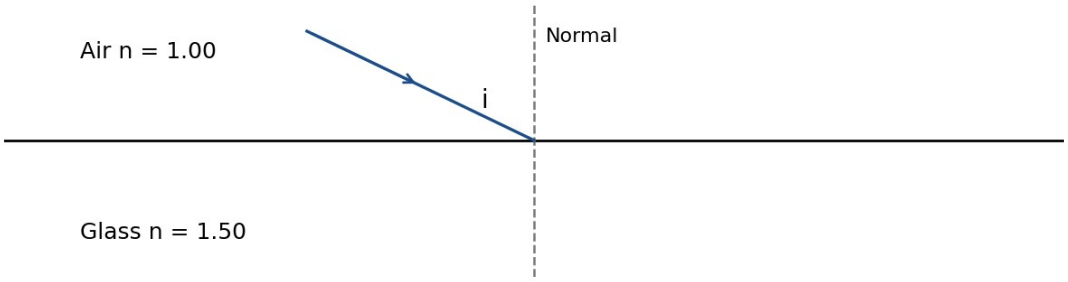
P22. A medium has refractive index 1.50. Calculate light speed in it. Use $c = 3.00 \times 10^8$ m/s.

P23. A medium has refractive index 1.60. Calculate light speed in it. Use $c = 3.00 \times 10^8$ m/s.

P24. Light of wavelength 600 nm in air enters glass of refractive index 1.50. Find its wavelength in glass.

P25. Light of wavelength 650 nm in air enters glass of refractive index 1.50. Find its wavelength in glass.

P26. Draw a ray entering glass obliquely from air. Label incident ray, refracted ray, boundary, normal, i and r.



P27. For glass of index 1.50 into air, compare incidence angles 40° and 50°. Which gives total internal reflection?

P28. Can light change speed at a boundary without changing direction? Explain.

P29. Why does white light split into colours through a prism?

P30. Design a method to estimate glass refractive index using a ray box.